

Research Paper

Submitted in Partial Fulfilment of the Requirements for the Financial Analyst
Internship Project

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Project Title:

Customer Segmentation & Churn Pattern Analytics in European Banking

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1. Introduction

Customer churn is an important challenge for banking institutions because the loss of customers can affect long-term customer relationships and business performance. Understanding which groups of customers are more likely to leave can help banks identify areas that require closer monitoring and develop more targeted retention strategies.

This project, “**Customer Segmentation & Churn Pattern Analytics in European Banking**,” focuses on analysing customer characteristics and identifying patterns associated with churn. The analysis uses a European banking customer dataset containing information such as geography, age, credit score, tenure, balance, number of products, credit card ownership, activity status and estimated salary. The target variable, **Exited**, indicates whether a customer has churned.

The project applies customer segmentation across geography, age, credit score, tenure and balance to understand how churn varies between different customer groups. It also examines customer engagement, financial characteristics and high-value customers to identify important patterns in the dataset.

The analysis is supported by exploratory data analysis and an interactive Streamlit dashboard. The dashboard allows users to apply filters and examine churn patterns across different customer segments. The objective is to convert the available customer data into clear analytical insights that can support data-driven customer retention planning.

2. Objectives

The main objective of this project is to analyse customer churn patterns in the European banking dataset and identify customer segments where churn is more concentrated. The analysis focuses on understanding differences in churn across demographic, geographic, financial and engagement-related characteristics.

The specific objectives are:

1. Measure the overall customer churn rate in the dataset.
2. Identify churn patterns across customer segments, including geography, age, credit score, tenure and balance.
3. Compare churn across European regions, particularly France, Germany and Spain.
4. Analyse high-value customer churn by examining customers with higher account balances.
5. Study customer engagement patterns by comparing active and inactive members.
6. Examine the relationship between customer characteristics and churn, including gender, number of products, credit card ownership and financial profile.
7. Identify segments with relatively higher observed churn rates that may require closer monitoring.
8. Develop an interactive Streamlit dashboard that allows users to explore churn patterns using different customer-segment filters. The project requirements specifically call for dynamic KPIs and drill-down views in the dashboard.

The overall purpose is to convert customer-level data into understandable analytical insights that can support data-driven retention planning.

3. Dataset Description

The project uses a European banking customer dataset containing 10,000 customer records. The dataset includes demographic, geographic, financial, product and engagement-related information. The target variable is `Exited`, which indicates whether a customer has churned. The dataset covers customers from France, Germany and Spain.

The original dataset contains the following variables:

Variable	Description
Year	Year associated with the customer record

Variable	Description
CustomerId	Unique identification number of the customer
Surname	Customer surname
CreditScore	Credit score of the customer
Geography	Country of the customer: France, Germany or Spain
Gender	Gender of the customer
Age	Age of the customer
Tenure	Number of years the customer has been associated with the bank
Balance	Customer's account balance
NumOfProducts	Number of banking products held by the customer
HasCrCard	Indicates whether the customer has a credit card
IsActiveMember	Indicates whether the customer is an active bank member
EstimatedSalary	Estimated salary of the customer
Exited	Churn indicator; 1 represents a churned customer and 0 represents a retained customer

The project requirements identify these variables as the primary fields for analysing customer segmentation and churn patterns.

Dataset characteristics

The dataset contains:

- **Total records:** 10,000
- **Total original columns:** 14
- **Geographies:** France, Germany and Spain
- **Target variable:** Exited
- **Target type:** Binary
- **Churned customers:** 2,037
- **Retained customers:** 7,963

- **Overall observed churn rate:** 20.37%

For the analytical dataset, the Year column was removed because all records belonged to the same year, while Surname was removed because it does not provide an analytical customer-segmentation variable for this project. The remaining variables were used for exploratory analysis and segmentation.

Segmentation variables

Following the project requirements, four derived segmentation variables were created:

Segmentation	Categories
Age Segment	<30, 30–45, 46–60, 60+
Credit Score Band	Low, Medium, High
Tenure Group	New, Mid-term, Long-term
Balance Segment	Zero-balance, Low-balance, High-balance

These categories were used to compare churn rates across different customer groups.

4. Data Preparation & Methodology

The analysis followed a structured data preparation and exploratory analysis process. The methodology was designed according to the project requirements, which include validating the dataset, removing non-analytical fields, creating customer segments and analysing churn across those segments.

4.1 Data Loading and Validation

The dataset was loaded into Python using the Pandas library. Initial checks were performed to understand the structure and quality of the data.

The following checks were conducted:

- Number of rows and columns
- Data types of variables
- Missing values
- Duplicate records
- Duplicate customer IDs

- Unique values of categorical variables
- Validity of binary variables such as HasCrCard, IsActiveMember and Exited
- Descriptive statistics for numerical variables

The dataset contained 10,000 records, and the data-quality checks did not identify missing values or duplicate customer IDs.

4.2 Removal of Non-Analytical Columns

Two columns were excluded from the analytical dataset:

- **Year** — all records belonged to the same year and therefore did not provide variation for the analysis.
- **Surname** — the customer's surname was not required for customer segmentation or churn analysis.

This resulted in a cleaner dataset focused on variables relevant to the project's analytical objectives.

4.3 Customer Segmentation

Derived segmentation variables were created to support granular churn analysis.

Age segmentation:

- <30
- 30–45
- 46–60
- 60+

Credit score segmentation:

- Low
- Medium
- High

Tenure segmentation:

- New
- Mid-term
- Long-term

Balance segmentation:

- Zero-balance
- Low-balance
- High-balance

These categories follow the segmentation framework specified in the project requirements.

4.4 Churn Rate Calculation

The main target variable was Exited, where:

- 1 = Customer churned
- 0 = Customer retained

The overall churn rate and segment-level churn rates were calculated using the proportion of customers with Exited = 1.

For example:

$$\text{Churn Rate} = (\text{Number of Churned Customers} / \text{Total Customers}) \times 100$$

This measure was applied across geography, age, tenure, credit score and balance segments.

4.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed using Pandas, Matplotlib and Seaborn.

The analysis examined:

- Overall churn
- Geographic differences
- Age-based churn
- Tenure patterns
- Credit score patterns
- Balance segments
- Gender differences
- Geography × Age interaction

- Customer engagement
- Number of products
- Credit card ownership
- Financial characteristics
- High-value customer churn

The project requirements specifically call for analysis of churn distribution, gender differences, geography-age interaction and financial stability, along with high-value customer analysis.

4.6 KPI Development

Four major analytical KPIs were calculated:

1. **Overall Churn Rate**
2. **High-Value Churn Ratio**
3. **Geographic Risk Index**
4. **Engagement Drop Indicator**

These KPIs were incorporated into the Streamlit dashboard to provide a concise view of the major churn patterns. The KPI framework is based on the project requirements.

4.7 Dashboard Development

An interactive Streamlit dashboard was developed to present the analysis. Users can filter the dataset by customer segments and examine the resulting changes in KPIs and analytical views.

The dashboard includes:

- Overall churn summary
- Segment analysis
- High-value customer analysis
- Engagement analysis
- Financial analysis
- Dynamic filters
- KPI updates

- Drill-down views
- Filtered-data download functionality

These features align with the dashboard requirements specified for the project.

Methodological Limitation

The analysis identifies **observed associations** between customer characteristics and churn. It does not establish that any particular characteristic directly causes customers to leave the bank. Therefore, the results should be interpreted as analytical patterns that can support further investigation and retention planning.

5. Exploratory Data Analysis

5.1 Overall Customer Churn

The first step of the analysis was to determine the overall level of customer churn in the dataset. Out of 10,000 customers, 2,037 customers had exited, while 7,963 customers remained with the bank.

The overall observed churn rate was therefore:

Overall Churn Rate = 20.37%



Figure 1: Overall Customer Churn Distribution

The chart shows that 20.37% of customers in the dataset have exited, while 79.63% have been retained. This indicates that customer churn represents a significant proportion of the analysed customer base.

This indicates that approximately one out of every five customers in the dataset had churned. The overall churn rate provides a baseline against which the churn rates of different customer segments can be compared.

The analysis shows that churn is not evenly distributed across all customer groups. Differences can be observed based on geography, age, balance, engagement and other customer characteristics. Therefore, segment-level analysis is important for understanding where churn is more concentrated.

5.2 Churn by Geography

The dataset contains customers from **France, Germany and Spain**. Churn rates were calculated separately for each country.

Geography	Total Customers	Churned Customers	Churn Rate
France	5,014	810	16.15%
Germany	2,509	814	32.44%
Spain	2,477	413	16.67%

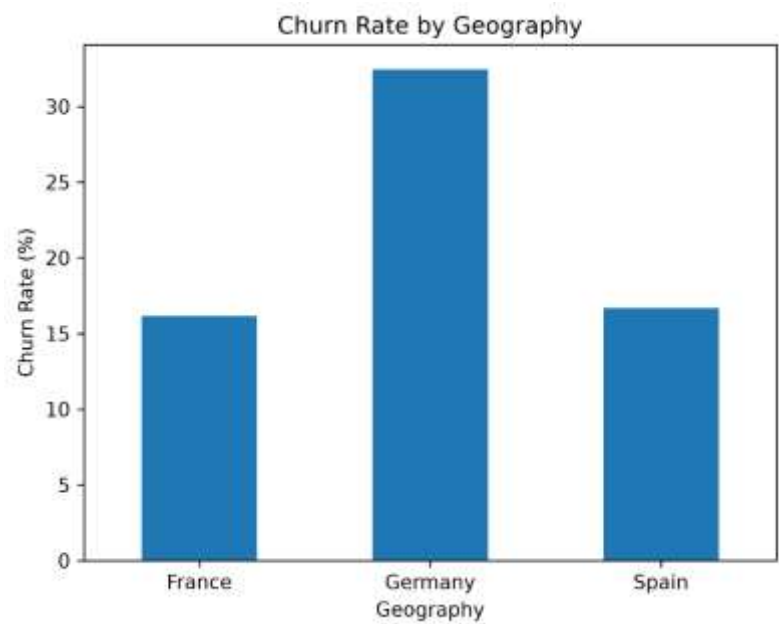


Figure 2: Customer Churn Rate by Geography

The chart shows noticeable differences in churn across the three countries. Germany records the highest observed churn rate at 32.44%, while France and Spain have lower rates of 16.15% and 16.67%, respectively. This indicates that geographic location is associated with substantial differences in observed customer churn within the dataset.

Germany recorded the highest observed churn rate at 32.44%. France recorded 16.15%, while Spain recorded 16.67%.

The difference indicates that churn is distributed differently across the three geographic segments. Germany's observed churn rate is approximately twice that of France and Spain in this dataset.

The geographic difference should be treated as an observed association rather than evidence that geography itself causes customer churn. Further investigation into customer experience, product usage and service patterns would be required to understand the reasons behind this difference.

5.3 Churn by Age

Customers were divided into four age groups according to the project-defined segmentation framework: <30, 30–45, 46–60 and 60+

Age Segment	Total Customers	Churned Customers	Churn Rate
<30	1,641	124	7.56%
30–45	6,248	956	15.30%
46–60	1,647	842	51.12%
60+	464	115	24.78%

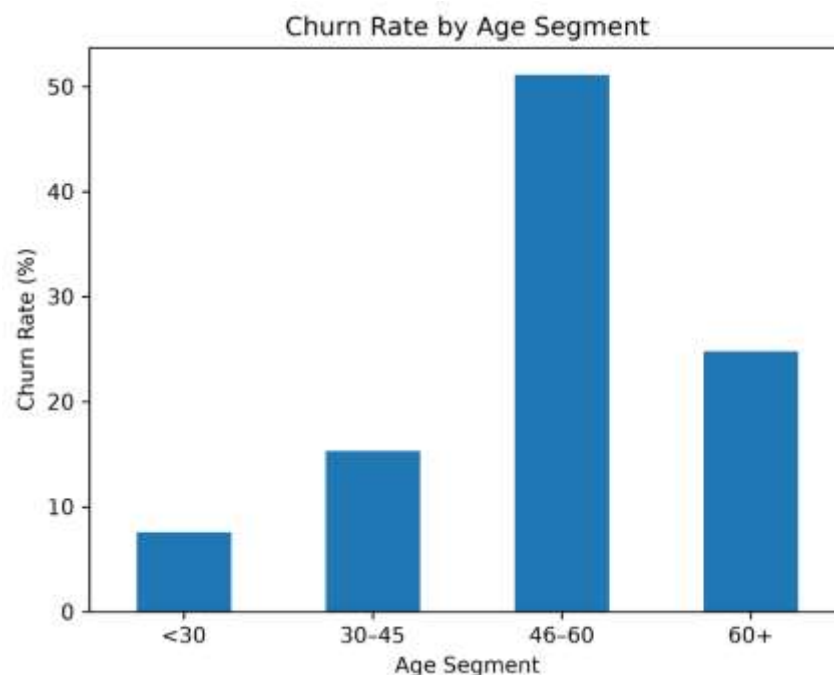


Figure 3: Customer Churn Rate by Age Segment

The chart shows a substantial difference in churn across age groups. Customers aged 46–60 have the highest observed churn rate at 51.12%, while customers below 30 have the lowest rate at 7.56%. The results indicate that age is an important segmentation dimension for understanding differences in observed churn.

The 46–60 age segment recorded the highest observed churn rate at 51.12%. This was substantially higher than the overall churn rate of 20.37%.

Customers below 30 had the lowest observed churn rate at 7.56%, while the 30–45 group recorded 15.30%. The 60+ segment had a churn rate of 24.78%.

These results indicate a notable concentration of churn among customers aged 46–60 in this dataset. This segment may therefore require closer examination of customer engagement, product usage and service experience.

5.4 Churn by Tenure

Customers were classified into three tenure groups: New, Mid-term and Long-term.

Tenure Group	Total Customers	Churned Customers	Churn Rate
New	3,505	741	21.14%
Mid-term	2,968	608	20.49%
Long-term	3,527	688	19.51%

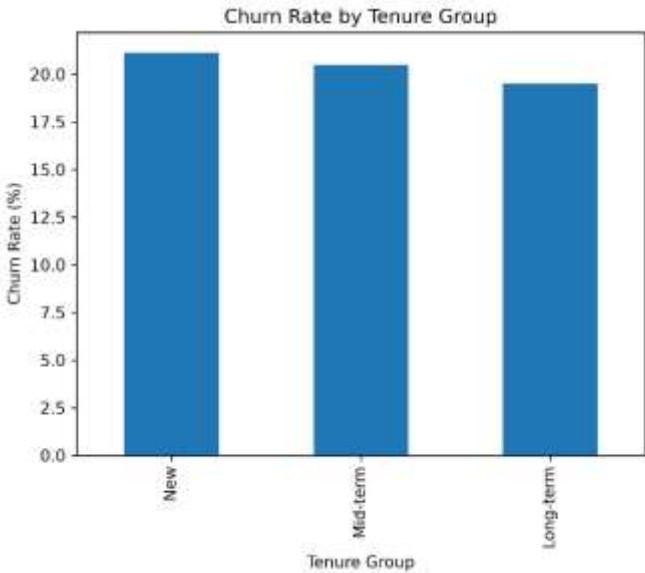


Figure 4: Customer Churn Rate by Tenure Group

The chart shows that churn rates are relatively similar across the three tenure groups. New customers have the highest observed churn rate at 21.14%, followed by mid-term customers at 20.49% and long-term customers at 19.51%. The relatively

small difference suggests that tenure alone does not show a large variation in observed churn within this dataset.

The New customer segment recorded the highest observed churn rate at 21.14%, followed closely by Mid-term customers at 20.49%. Long-term customers had the lowest rate at 19.51%.

Although the differences are relatively small, the results suggest that early-stage customer relationships should be monitored as part of the bank's retention analysis.

5.5 Churn by Credit Score

Customers were divided into three credit score categories: **Low, Medium and High**. The segmentation was used to examine whether churn rates differed across credit-score groups.

Credit Score Band	Total Customers	Churned Customers	Churn Rate
Low	3,034	660	21.75%
Medium	3,818	753	19.72%
High	3,148	624	19.82%

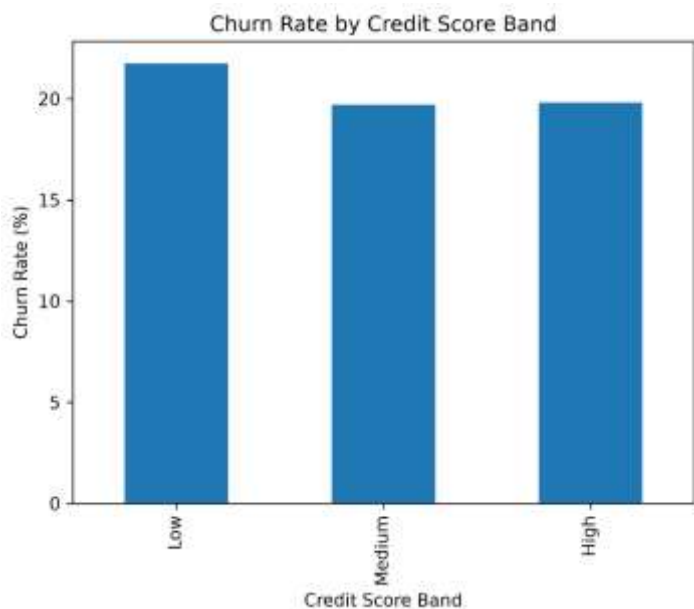


Figure 5: Customer Churn Rate by Credit Score Band

The chart shows that customers in the Low credit score band have the highest observed churn rate at 21.75%. Medium and High credit score groups have lower and relatively similar churn rates of 19.72% and 19.82%, respectively. This suggests

that lower credit scores are associated with somewhat higher observed churn in the dataset.

The Low credit score segment recorded the highest observed churn rate at 21.75%. The Medium and High segments recorded churn rates of 19.72% and 19.82%, respectively.

The difference between the groups is relatively moderate. Therefore, credit score appears to show a smaller variation in observed churn compared with some of the other segments, such as geography and age.

These results can be used to identify customer groups for further investigation, but they should not be interpreted as evidence that credit score directly causes churn.

5.6 Churn by Balance Segment

Customer balances were divided into three groups: Zero-balance, Low-balance and High-balance. The High-balance category represents customers whose balances were above the dataset median.

Balance Segment	Total Customers	Churned Customers	Churn Rate
Zero-balance	3,617	500	13.82%
Low-balance	1,383	288	20.82%
High-balance	5,000	1,249	24.98%

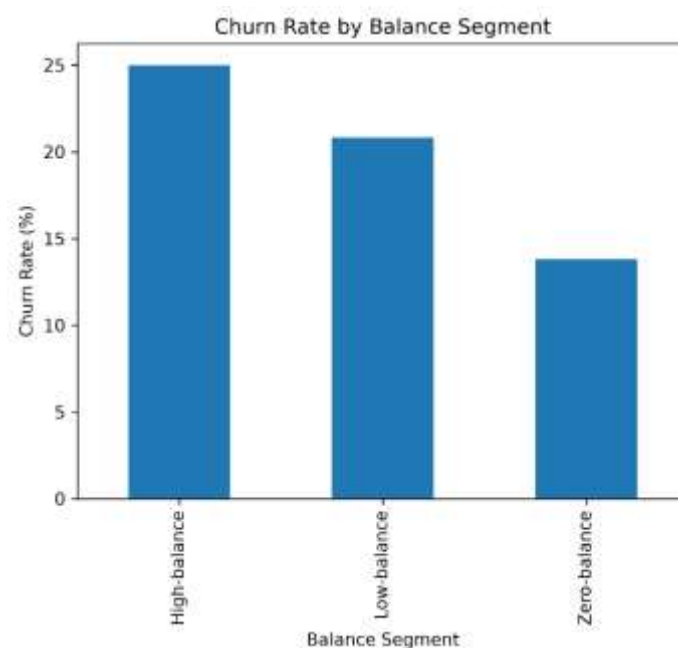


Figure 6: Customer Churn Rate by Balance Segment

The chart shows higher observed churn among customers with higher balances. High-balance customers have a churn rate of 24.98%, compared with 20.82% for low-balance customers and 13.82% for zero-balance customers. This indicates that balance level is associated with meaningful differences in observed churn and can be useful for identifying financially exposed customer segments.

The High-balance segment recorded the highest observed churn rate at 24.98%, compared with 20.82% for Low-balance customers and 13.82% for Zero-balance customers.

This finding is particularly relevant to the project's high-value customer analysis because customers with higher balances represent an important customer group for monitoring. The higher observed churn rate indicates greater balance exposure among churned customers, although the analysis does not establish that balance itself causes churn.

5.7 Churn by Gender

Gender differences were also examined to understand whether the observed churn rate varied between male and female customers.

The analysis compares the proportion of customers with Exited = 1 within each gender group.

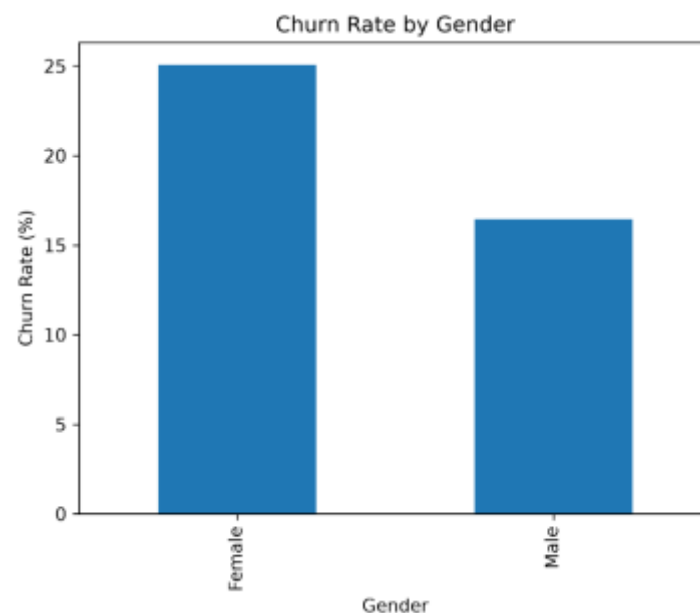


Figure 7: Churn Rate by Gender

The gender analysis provides an additional demographic perspective on customer churn and can be considered alongside the geographic and age-based analysis. However, gender should be interpreted as a descriptive segmentation variable rather than as a causal explanation for customer churn.

5.8 Geography and Age Interaction

The project also examined the interaction between Geography and Age to identify whether age-related churn patterns differed across countries.

This analysis is important because the overall age pattern and geographic pattern may not be identical within each country. A customer segment can have a different churn profile when analysed jointly with another characteristic.

The geography × Age analysis therefore provides a more granular view of customer churn and helps identify combinations of customer characteristics that may require closer investigation.

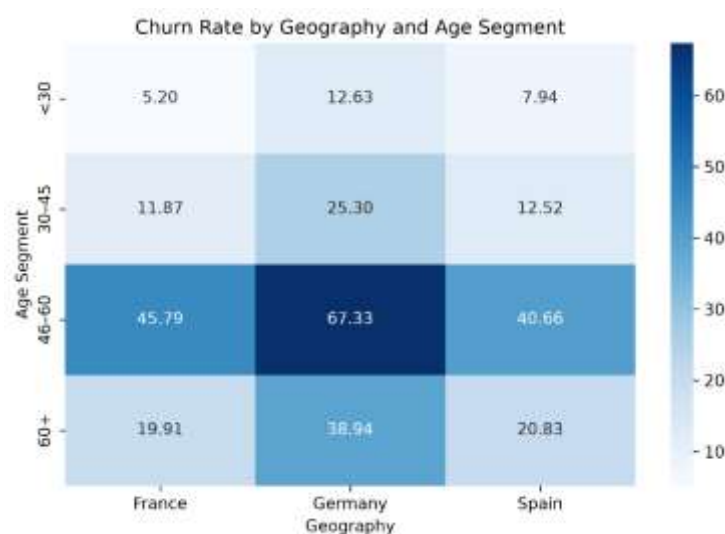


Figure 8: Churn Rate by Geography and Age Segment

The heatmap highlights differences in observed churn across combinations of geography and age. It shows that churn varies not only between countries but also across age segments within each country. This two-dimensional view helps identify customer groups where geographic and demographic characteristics coincide with higher observed churn.

The results were visualised using a heatmap in the exploratory analysis and incorporated into the Streamlit dashboard.

5.9 Customer Engagement and Churn

Customer engagement was analysed using the IsActiveMember variable, which indicates whether a customer is an active member of the bank.

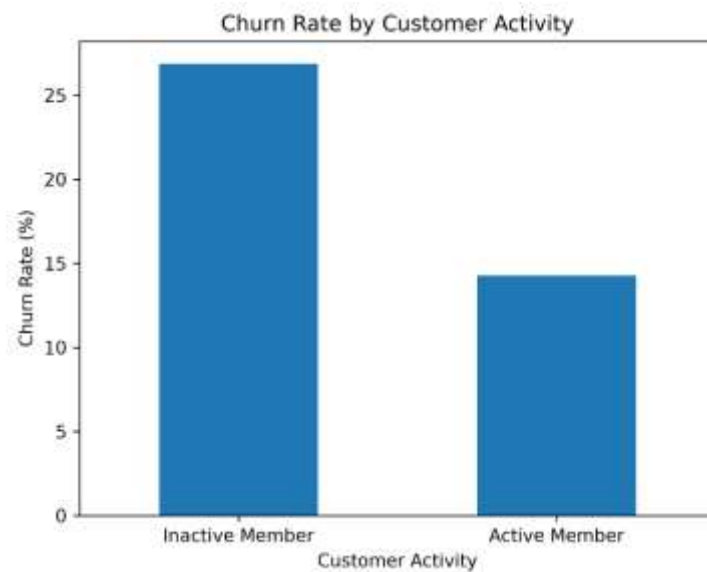


Figure 9: Churn Rate by Customer Activity

The analysis showed a clear difference in observed churn between active and inactive customers. The Engagement Drop Indicator was 12.58 percentage points, calculated as the difference between the observed churn rates of inactive and active customers.

This indicates that inactive customers had a substantially higher observed churn rate than active customers in the dataset.

Customer engagement is therefore an important dimension for further investigation. Monitoring inactive customers and examining their product usage and service interactions may help identify customers who require additional attention.

However, the observed relationship should not be interpreted as proof that inactivity directly causes churn.

5.10 Number of Products and Churn

The analysis also examined the relationship between the number of banking products held by customers and their churn status.

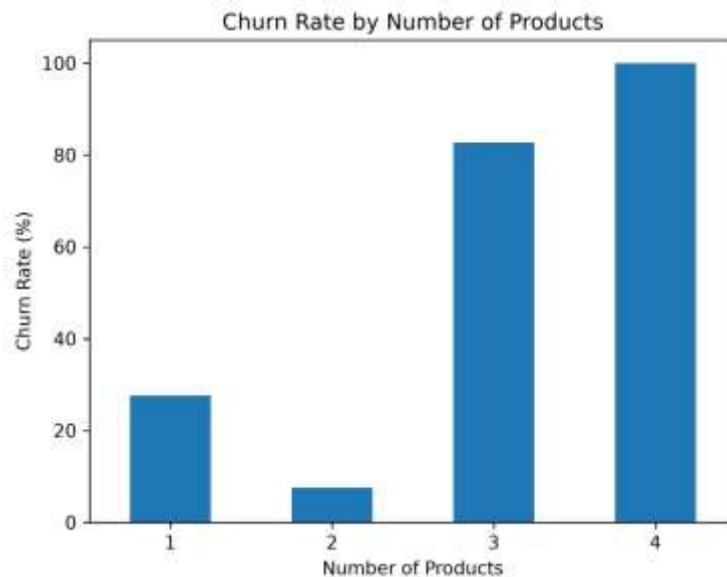


Figure 10: Churn Rate by Number of Products

The NumOfProducts variable contains values ranging from 1 to 4 products. Comparing churn across these groups provides an indication of whether customer product relationships differ between churned and retained customers.

This analysis is useful because customers with different numbers of products may have different levels of interaction with the bank. The results were visualised in the exploratory analysis and incorporated into the dashboard's Engagement section.

The number of products should be considered together with other customer characteristics rather than being treated as an independent explanation for churn.

5.11 Credit Card Ownership and Churn

Credit card ownership was analysed using the HasCrCard variable.

Customers were divided into two groups:

- Customers with a credit card
- Customers without a credit card

The purpose of this analysis was to determine whether observed churn patterns differed according to credit card ownership.

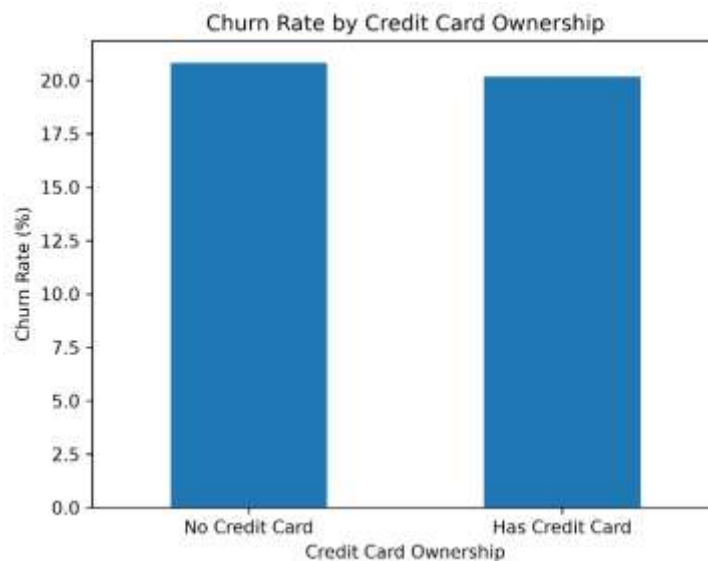


Figure 11: Churn Rate by Credit Card Ownership

This provides an additional view of the customer's relationship with the bank and complements the analysis of the number of products and customer engagement.

5.12 Financial Profile of Churned and Retained Customers

The financial profile analysis compared Balance and Estimated Salary between churned and retained customers.

Balance was particularly important because the project includes a dedicated high-value customer analysis. Customers with higher balances were classified as High-balance customers using the median balance as the threshold.

The analysis also used a salary-versus-balance scatter plot to examine how these two financial variables were distributed across customers with different churn statuses.

This analysis helps identify whether churn is concentrated among customers with relatively higher financial exposure. However, balance and salary are treated as descriptive characteristics in this project and not as direct measures of revenue loss.

5.13 Overall EDA Summary

The exploratory analysis identified several notable differences in observed churn across customer segments.

The most prominent patterns were:

- Overall observed churn rate: **20.37%**
- Germany churn rate: **32.44%**

- Age 46–60 churn rate: **51.12%**
- High-balance churn rate: **24.98%**
- Engagement Drop Indicator: **12.58 percentage points**
- New customer churn rate: **21.14%**
- Low credit-score churn rate: **21.75%**

These findings indicate that churn is not evenly distributed across the customer base. Certain demographic, geographic, financial and engagement segments show higher observed churn rates and therefore provide useful areas for further investigation.

The findings also support the project's objective of identifying granular customer segments rather than relying only on the overall churn rate.

6. High-Value Customer Analysis

6.1 Definition of High-Value Customers

For this analysis, customers were segmented according to their account balance. Customers with balances above the dataset median were classified as High-balance customers.

The dataset contained 5,000 high-balance customers. Among these customers, 1,249 had churned, resulting in an observed churn rate of:

High-Value Churn Ratio = 24.98%

This is higher than the overall customer churn rate of 20.37%.

The comparison indicates that the high-balance segment had a higher observed churn rate than the overall customer population.

6.2 High-Value Customer Churn by Geography

The high-balance customer group was further examined by geography to identify differences across France, Germany and Spain.

This analysis is useful because the overall geographic analysis showed substantial differences in observed churn, and combining geography with the high-balance segment provides a more granular view of customer risk.

The analysis showed that high-value customer churn varied across the three countries. Germany was an important area for further investigation because it also recorded the highest overall geographic churn rate of 32.44%.

The combination of geography and balance therefore provides a useful perspective for identifying customer groups that may warrant closer monitoring.

6.3 High-Value Customer Churn by Age

High-balance customers were also analysed across the four age segments.

The analysis showed that age-related churn patterns remained relevant within the high-value customer population. In particular, the 46–60 age group was an important segment to examine because this group recorded the highest overall observed churn rate of 51.12%.

This suggests that combining customer value with demographic segmentation can provide more detailed information than analysing either characteristic separately.

6.4 High-Value Customer Churn by Tenure

Tenure was analysed among high-balance customers to determine whether churn patterns differed between New, Mid-term and Long-term customers.

The tenure analysis provides additional context for understanding the customer relationship lifecycle. New customers can be examined from an onboarding and early-engagement perspective, while long-term customers can be examined in terms of relationship continuity and engagement.

The overall dataset showed observed churn rates of:

Tenure Group	Churn Rate
New	21.14%
Mid-term	20.49%
Long-term	19.51%

The differences are relatively moderate, suggesting that tenure alone does not explain the variation in churn observed in the dataset.

6.5 High-Value Customer Churn by Credit Score

Credit score was also examined within the broader high-value customer analysis.

Across the complete dataset, the observed churn rates were:

Credit Score Band	Churn Rate
Low	21.75%
Medium	19.72%
High	19.82%

The Low credit-score group had the highest observed churn rate among the three credit-score categories.

This analysis provides an additional financial-risk dimension when studying high-balance customers. However, credit score and balance should be interpreted as customer characteristics associated with observed churn rather than as direct causes.

6.6 Salary and Balance Relationship

The project also examined the relationship between Estimated Salary and Balance using a scatter plot.

The purpose was to understand whether customers with higher balances also consistently had higher estimated salaries and whether churned customers were concentrated within particular financial ranges.

The analysis provides a visual understanding of the financial distribution of customers and helps distinguish between balance exposure and salary characteristics.

Importantly, the analysis does not treat customer balance as equivalent to bank revenue or financial loss. Instead, balance is used as a proxy for customer financial exposure within the dataset.

6.7 High-Value Customer Risk Interpretation

The high-value analysis provides three important observations:

1. High-balance customers had an observed churn rate of 24.98%, compared with 20.37% overall.
2. Geographic and demographic segmentation can reveal additional patterns within the high-balance population.
3. High-balance customers who churn represent a customer group that may deserve closer monitoring because of their higher account-balance exposure.

The findings support a more targeted approach to churn monitoring in which customer value is considered alongside geography, age, engagement and other characteristics.

The project requirements specifically call for analysis of high-balance churners and salary-versus-balance patterns to understand potential financial risk.

7. Key Findings

The analysis of 10,000 European banking customer records identified several important patterns in customer churn. The findings below summarise the major observations from the exploratory and high-value customer analyses.

7.1 Overall Churn

The dataset recorded an overall observed churn rate of 20.37%, with 2,037 out of 10,000 customers classified as churned.

This provides the baseline for comparing churn across different customer segments.

7.2 Geographic Differences

Churn varied considerably across the three countries.

- Germany: 32.44%
- Spain: 16.67%
- France: 16.15%

Germany therefore showed a substantially higher observed churn rate than the other two geographic segments in this dataset.

7.3 Age-Based Churn

The strongest age-related pattern was observed among customers aged 46–60, where the churn rate was 51.12%.

In comparison:

- Customers below 30: 7.56%
- Customers aged 30–45: 15.30%
- Customers aged 60+: 24.78%

This makes the 46–60 segment an important group for further investigation.

7.4 High-Balance Customers

High-balance customers had an observed churn rate of 24.98%, compared with the overall churn rate of 20.37%.

The result indicates that churn was relatively more concentrated among customers with higher account balances.

7.5 Customer Engagement

The Engagement Drop Indicator was 12.58 percentage points, representing the difference between the observed churn rates of inactive and active customers.

This indicates a substantial difference in churn patterns between the two engagement groups.

7.6 Tenure

Tenure showed comparatively smaller differences:

- New: 21.14%
- Mid-term: 20.49%
- Long-term: 19.51%

The New customer segment had the highest observed churn rate, although the difference from the other tenure groups was relatively modest.

7.7 Credit Score

The Low credit-score segment recorded the highest observed churn rate at 21.75%.

Medium and High credit-score segments recorded 19.72% and 19.82%, respectively.

The relatively small differences suggest that credit score showed less variation in observed churn than age or geography.

7.8 Key Analytical Interpretation

Overall, the analysis demonstrates that customer churn is not evenly distributed across the customer base. The most notable variations were observed across geography, age, balance and engagement.

The combination of these dimensions is more informative than examining any single variable independently. For example, combining customer value with geographic or demographic segmentation can help identify more specific groups for further investigation.

These findings should be interpreted as observed associations within the dataset rather than causal relationships.

Key Findings Summary Table:

Indicator	Observed Result
Overall Churn Rate	20.37%
Germany Churn Rate	32.44%
Highest Age-Segment Churn	51.12% (Age 46–60)
High-Value Churn Ratio	24.98%
Engagement Drop Indicator	12.58 percentage points
Highest Tenure Churn	21.14% (New)
Highest Credit Score Churn	21.75% (Low)

8. Recommendations

The recommendations below are based on the observed churn patterns identified during the analysis. They are intended to support further investigation and targeted retention planning rather than assume that any single customer characteristic directly causes churn.

8.1 Investigate Country-Specific Churn Patterns

Germany recorded an observed churn rate of 32.44%, considerably higher than France at 16.15% and Spain at 16.67%.

The bank could therefore conduct a more detailed review of customers in Germany, focusing on areas such as:

- Customer service experience

- Product usage
- Customer engagement
- Number of products held
- Service interactions
- Other country-specific operational factors

The purpose should be to identify what additional factors may be associated with the higher churn pattern.

8.2 Focus on the 46–60 Age Segment

Customers aged 46–60 recorded a 51.12% observed churn rate, the highest among the age groups analysed.

This segment could be monitored more closely through:

- Targeted engagement programmes
- Regular customer feedback
- Product-use reviews
- Personalised communication
- Early identification of disengagement

These activities should be evaluated using customer-level evidence to determine whether they improve retention.

8.3 Monitor High-Balance Customers

High-balance customers recorded a 24.98% observed churn rate, higher than the overall rate of 20.37%.

Banks could therefore develop a high-value customer monitoring framework that considers:

- Account balance
- Customer activity
- Product relationships
- Tenure
- Geography
- Recent changes in engagement

This would allow high-balance customers showing multiple risk indicators to receive closer attention.

8.4 Strengthen Customer Engagement

The analysis identified an Engagement Drop Indicator of 12.58 percentage points between inactive and active customers.

Banks could monitor inactive customers more regularly and examine whether reduced engagement is accompanied by changes in product usage or other customer behaviours.

Potential approaches include:

- Relevant product communication
- Customer engagement campaigns
- Digital banking engagement
- Periodic customer check-ins
- Personalised service interactions

8.5 Improve New-Customer Onboarding

New customers had an observed churn rate of 21.14%, slightly higher than Mid-term customers at 20.49% and Long-term customers at 19.51%.

This suggests that onboarding and early-stage engagement should be included in the retention review.

Banks could examine:

- Onboarding experience
- Initial product adoption
- Early customer communication
- Digital service usage
- Customer support during the first stages of the relationship

8.6 Examine Low Credit-Score Customers

The Low credit-score group recorded an observed churn rate of 21.75%, compared with 19.72% for Medium and 19.82% for High credit-score customers.

Rather than treating credit score as a direct cause of churn, banks could investigate whether other customer characteristics or product/service patterns are contributing to the observed difference.

8.7 Use Multi-Dimensional Customer Segmentation

The analysis demonstrates that examining one variable at a time can hide more detailed customer patterns.

A more comprehensive monitoring approach could combine:

Geography + Age + Balance + Engagement + Tenure + Product Usage

For example, a high-balance customer in a higher-churn geographic or age segment who is also inactive could be flagged for further investigation.

This approach would allow retention teams to focus their analytical attention on specific customer profiles rather than applying identical strategies to the entire customer base.

8.8 Build Continuous Churn Monitoring

The Streamlit dashboard developed as part of this project can be used as an exploratory monitoring tool.

The dashboard allows users to:

- Filter customer segments
- Compare churn rates
- Examine high-value customers
- Analyse engagement
- Explore financial characteristics
- Download filtered customer data

The project requirements specifically call for dynamic filters, KPI updates and drill-down views, which are incorporated into the dashboard.

Recommendation Summary

Observed Pattern

Germany: 32.44% churn

Suggested Area of Action

Investigate country-specific factors

Observed Pattern	Suggested Area of Action
Age 46–60: 51.12% churn	Increase monitoring and engagement
High-balance: 24.98% churn	Monitor high-value customer segment
Engagement gap: 12.58 pp	Strengthen engagement monitoring
New customers: 21.14% churn	Review onboarding and early engagement
Low credit score: 21.75% churn	Investigate associated customer/product patterns

9. Conclusion

This project analysed customer segmentation and churn patterns using a dataset of 10,000 European banking customers from France, Germany and Spain. The analysis examined customer characteristics across geography, age, tenure, credit score, balance, engagement, product usage and other financial variables.

The overall observed customer churn rate was 20.37%. However, the analysis showed that churn was not evenly distributed across the customer base. Germany recorded an observed churn rate of 32.44%, while customers aged 46–60 recorded the highest age-segment churn rate of 51.12%.

The analysis also identified an important pattern among high-balance customers. This group had an observed churn rate of 24.98%, which was higher than the overall churn rate. This highlights the usefulness of combining customer value with demographic, geographic and engagement characteristics when analysing churn.

Customer engagement was another important dimension. The difference between observed churn rates for inactive and active customers resulted in an Engagement Drop Indicator of 12.58 percentage points. New customers and customers in the Low credit-score segment also showed somewhat higher observed churn rates than their respective comparison groups.

The project further demonstrated the value of multi-dimensional customer segmentation. Analysing characteristics such as geography, age, balance and engagement together provides a more detailed understanding of where churn is concentrated than relying only on the overall churn rate.

An interactive Streamlit dashboard was developed to make these findings easier to explore. The dashboard provides dynamic filters, KPI summaries, segment-level analysis, high-value customer analysis, engagement and financial views, as well as

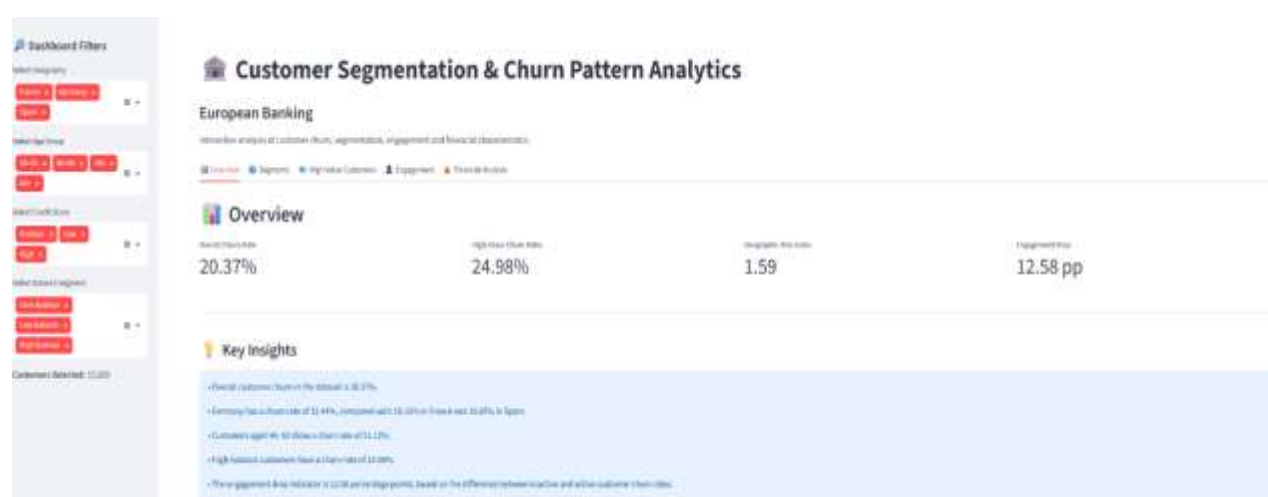
the ability to download filtered customer data. This supports the project's objective of presenting customer churn insights through an accessible analytical interface.

Overall, the project demonstrates how exploratory data analysis and customer segmentation can be used to identify meaningful churn patterns in banking data. The findings can support further investigation and help organisations develop more targeted customer-retention strategies. However, the results represent observed associations rather than causal relationships, and additional operational and customer-level information would be required to validate the findings before implementing specific interventions.

10. Dashboard Description

An interactive Streamlit dashboard was developed to present the customer churn analysis in an accessible and user-friendly format. The dashboard converts the analytical results into interactive visualisations and allows users to explore different customer segments using filters.

The dashboard is divided into five main sections.



10.1 Overview

The Overview section provides a summary of the selected customer population. It includes:

- Overall Churn Rate
- High-Value Churn Ratio
- Geographic Risk Index
- Engagement Drop Indicator

- Key Insights
- Project Information
- Dataset Preview
- Methodology Note
- Retention Recommendations
- Filtered customer data download

The KPIs update dynamically when the user changes the filters.

10.2 Segment Analysis

The Segments section provides detailed analysis of churn across:

- Geography
- Age
- Tenure
- Credit Score
- Balance
- Geography × Age

This section allows users to compare customer groups and identify differences in observed churn rates.

10.3 High-Value Customer Analysis

The High-Value Customers section focuses on customers classified as high-balance customers.

It provides information about:

- Number of high-value customers
- Number of high-value churners
- High-value churn rate
- Geographic distribution of high-value churn
- Customer-level high-value analysis

Customer Segmentation & Churn Pattern Analytics

European Banking

Interactive analysis of customer churn, segmentation, engagement and financial characteristics.

Overview Segments High-Value Customers Engagement Financial Analysis

High-Value Customer Analysis

High-Value Customers

5,000

High-Value Churners

1,249

High-Value Churn Rate

24.98%

High-Value Churn by Geography

Geography	High-Value Customers	Churned High-Value	Churn Rate
France	1,000	250	25.00%
Germany	1,500	375	25.00%
Spain	900	225	25.00%



This section supports the project's objective of examining churn among higher-balance customers.

10.4 Engagement Analysis

The Engagement section examines customer activity and product relationships.

It includes analysis of:

- Active versus inactive customers
- Churn patterns by customer activity
- Number of products
- Relationship between engagement and churn

This helps identify differences in observed churn between more and less active customer groups.

10.5 Financial Analysis

The Financial Analysis section examines the financial characteristics of churned and retained customers.

It includes:

- Average balance comparison

- Average estimated salary comparison
- Salary versus balance visualisation
- High-value customer financial profile



The section provides additional context for understanding the financial characteristics associated with customer churn.

10.6 Interactive Filters

The dashboard provides sidebar filters for:

- Geography
- Age Segment
- Credit Score
- Balance Segment

When a filter is changed, the dashboard dynamically updates the relevant KPIs, tables and visualisations.

This functionality allows users to perform drill-down analysis and examine specific customer populations, which is consistent with the project's requirement for dynamic filters and drill-down views.

10.7 Data Download

A download feature was also implemented to allow users to export the currently filtered customer dataset as a CSV file.

This provides a practical way for analysts to perform additional analysis outside the dashboard.

11. References

1. **Project Requirements Document:** *Customer Segmentation & Churn Pattern Analytics in European Banking*.
2. **European Banking Customer Dataset:** European_Bank.csv.
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4. **Matplotlib Documentation** — used for data visualisation.
5. **Seaborn Documentation** — used for statistical visualisation.
6. **Streamlit Documentation** — used for developing the interactive dashboard.